

AGENTR: Agentic Data Science with R

Full Program, October 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Agentic Data Science with R

Format: Online, cohort-based

Duration: 2 weeks, 8 sessions, 2 hours per session, 16 contact hours

Dates: 19/10/2026 – 01/11/2026

Days: Monday, Wednesday, Friday (18h00 to 20h00), Sunday (10h00 to 12h00)

Time: Morocco time

Price: 200€ (2000 MAD)

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1.1 Course Description

AGENTR is the capstone of the MHDSR pathway: a practical, hands-on introduction to *agentic data science*, using large language models (LLMs) and coding agents to build, run, and reproduce data analysis pipelines in R from the command line.

The course works through the full agentic workflow: choosing between frontier and local models, prompting and context engineering, calling R and shell tools from an agent via structured tool use, grounding output with retrieval over data and documents, packaging reusable skills and multi-step agent workflows, and evaluating agent output with tests, guardrails, and human review.

The emphasis throughout is on keeping the analyst in control: verification, provenance, cost, privacy, and honest reporting of AI-assisted work.

1.2 Target Audience

R users, data analysts, researchers, PhD students, and public health professionals who want to use LLMs and coding agents for real analysis work. No machine learning background required.

1.3 Prerequisites

Comfort with R and the command line, and basic data analysis experience (as covered in BIOSTATR), are recommended but not mandatory.

1.4 Learning Objectives

By the end of AGENTR, participants will be able to:

1. describe what agentic systems are, and where they help or fail in data science;
2. set up a CLI-based agentic workflow for R projects (API keys, local runtimes, project scaffolding);
3. compare frontier and local LLMs on capability, cost, latency, and privacy;
4. write effective prompts and manage context for data analysis tasks;
5. call R functions and external tools from a model using structured tool use / function calling;
6. build retrieval over datasets, codebases, and documents to ground model output;
7. package reusable skills and multi-step agent workflows for recurring analyses;
8. evaluate agent output with tests and human review, and add guardrails;
9. manage reproducibility, provenance, cost, and data governance of AI-assisted analyses;
10. deliver a reproducible, agent-assisted analysis pipeline with an audit trail.

2 Detailed Schedule

Table 1: AGENTR detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	19/10/2026 (Mon)	18h00- 20h00	Introduction to Agentic Data Science: LLMs, Agents & the CLI Workflow for R	What LLMs and agents are, chat vs agent vs pipeline, the terminal-first workflow, project scaffolding, when agentic help pays off and when it does not
2	1	21/10/2026 (Wed)	18h00- 20h00	Frontier vs Local Models: Access, Cost, Latency & Privacy	Hosted frontier APIs, local model runtimes, model families and sizes, context windows, token cost and latency, data-privacy trade-offs, choosing a model per task
3	1	23/10/2026 (Fri)	18h00- 20h00	Prompting and Context Engineering for Data Analysis	Task framing, few-shot examples, structured output, supplying data dictionaries and code context, context limits, prompt iteration and versioning
4	1	25/10/2026 (Sun)	10h00- 12h00	Tool Use and Function Calling: Driving R and Shell Tools from an Agent	Structured tool/function calling, exposing R functions and shell commands as tools, the read-eval-act loop, sandboxing, approval gates, failure handling
5	2	26/10/2026 (Mon)	18h00- 20h00	Retrieval-Augmented Workflows over Data, Code & Literature	Chunking and embeddings, vector stores, retrieval over project files and datasets, grounding answers in sources, citations, keeping retrieval reproducible

No.	Week	Date	Time	Topic	Main Content
6	2	28/10/2026 (Wed)	18h00- 20h00	Agent Skills and Reusable Multi-Step Pipelines from the Command Line	Packaging recurring analyses as skills, multi-step and multi-agent workflows, parameterized runs, scheduling, chaining agents into a data pipeline
7	2	30/10/2026 (Fri)	18h00- 20h00	Evaluation, Guardrails, Provenance & Reproducibility of Agent-Assisted Work	Test-based checks on agent output, human review, guardrails and policies, logging prompts and responses, cost tracking, provenance and audit trails, honest reporting
8	2	01/11/2026 (Sun)	10h00- 12h00	End-to-End Agentic Analysis Project	Full workflow: scaffold, retrieve, prompt, run tools, evaluate, and report a reproducible agent-assisted analysis with an audit trail

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, prompts, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

Each course ends with a short reproducible Quarto report or mini-project, presented and discussed in Session 8. For AGENTR, this is a reproducible agent-assisted analysis pipeline with an audit trail.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.

Software: R (≥ 4.4) and RStudio required; a terminal, an LLM CLI / coding agent, and API access and/or a local model runtime are used throughout.

Attendance: Full attendance at all 8 sessions is recommended given the cumulative structure.